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Review

A Survey on Surface Defect Inspection Based on Generative Models in Manufacturing

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Abstract: Surface defect inspection based on deep learning has demonstrated outstanding performance in improving detection accuracy and model generalization. However, the small scale of defect datasets always limits the application of deep models in industry. Generative models can obtain realistic samples in a very cheap way, which can effectively solve this problem and thus has received widespread attention in recent years. This paper provides a comprehensive analysis and summary of the current studies of surface defect inspection methods proposed between 2022 and 2024. First, according to the use of generative models, these methods are classified into four categories: Variational Auto-Encoders (VAEs), Generative Adversarial Networks (GANs), Diffusion Models (DMs), and multi-models. Second, the research status of surface defect inspection based on generative models in recent years is discussed from four aspects: sample generation, detection objective, inspection task, and learning model. Then, the public datasets and evaluation metrics that are commonly used for surface defect inspection are discussed, and a comparative evaluation of defect inspection methods based on generative models is provided. Finally, this study discusses the existing challenges for the defect inspection methods based on generative models, providing insights for future research.

Keywords: deep learning; generative models; surface defect inspection; survey

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Citation: He, Y.; Li, S.; Wen, X.; Xu, J. A Survey on Surface Defect Inspection Based on Generative Models in Manufacturing. *Appl. Sci.* **2024**, *14*, 6774. https://doi.org/10.3390/app14156774

Academic Editor: Sungho Kim

Received: 16 July 2024
Revised: 31 July 2024
Accepted: 31 July 2024
Published: 2 August 2024

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1. Introduction

The manufacturing industry is the foundation of national economic development, directly reflecting the productivity level and international competitiveness of this country. To maintain competitiveness in the international market, excellent product quality is key. In the actual production process, it is inevitable that various defects appear on the surface of the product due to the difficulty in controlling the quality of raw materials, complex production technology, etc. Therefore, surface defect inspection, as an important component of quality production, plays a crucial role.

Generally, surface defects can be defined as the areas with local texture colors or structures that are inconsistent with the normal background surfaces [1]. According to the difference in material stiffness of a product, there are two types of surface defects in industry, i.e., the defects on rigid material surfaces, e.g., metal, wood, and ceramics, etc., and the defects on flexible material surfaces, e.g., fabric, leather, and rubber, etc. [2]. Visually, the former show irregular cracks, chipping inclusions, and stripe scratches, whereas the latter present bubbles, and dot or short-line type defects. Figure 1 shows the examples of two types of surface defects. Regardless of either type of surface defects, they not only influence the appearance of a product but also always become the starting point of physical damage or chemical corrosion, which can affect its quality and performance [3]. However, traditional defect inspection is performed by human vision, which is time-consuming,

hard sledding, and inefficient [4]. The studies [5–7] reported the drawbacks of the manual inspection in details. It is necessary to develop an automatic defect inspection system to replace the unreliable manual inspection.

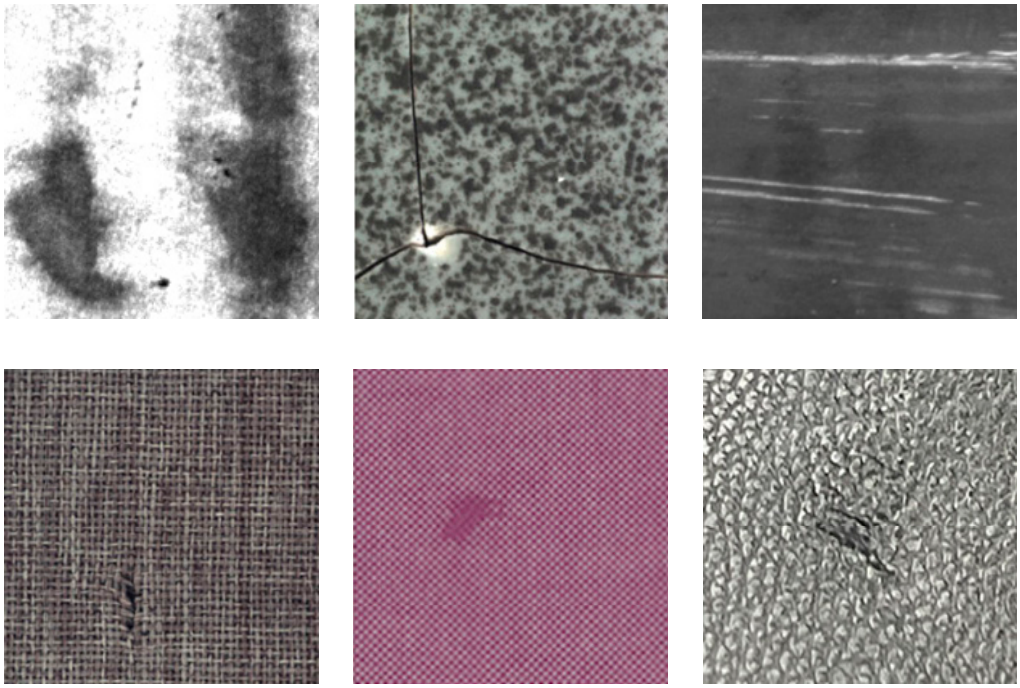

Figure 1. Examples of surface defect images. Upper row: examples of defects on rigid surfaces (https://pan.baidu.com/s/1l_RjTP7aTwr57ahcwelTpA, accessed on 25 July 2024). Bottom row: examples of defects on flexible surfaces (https://ibug.doc.ic.ac.uk/resources/fabrics/, accessed on 25 July 2024).

To achieve this, automatic defect inspection schemes were proposed in the 1980s [8]. Continuously, with the wide applications of vision sensors, more and more inspection methods based on machine learning (ML) and image processing techniques have emerged, mainly during the period 1981–2010. Usually, these methods extract well-designed handcraft features and then feed them into a pattern recognizer, hence also known as feature engineering. In past decades, related works have been reported. The surveys [9–11] summarized the surface defect inspection methods in the 1980s, 1990s, and 2000s, respectively. Commonly, the traditional ML-based methods can be divided into four types: statistical methods, structural methods, filter-based methods, and model-based methods. However, no matter which kind of method of those mentioned above, the key to success is to design appropriate handcraft features, resulting in the inspection system having poor generalization and universality.

In 2012, AlexNet [12] achieved great success in the visual recognition competition, attracting the world's attention to the deep learning (DL) technique, especially the convolutional neural network (CNN). These models adopt neural networks with multiple convolutional layers to learn the high-level features with the powerful knowledge representation ability. In various computer vision tasks, e.g., image classification and object detection, a CNN has shown higher accuracy, better generalization, and less sensitivity to application conditions than traditional ML methods [13–15]. It is natural that such models are employed in surface defect inspection and have become the most popular tools until now. Understandably, there are many reviews, such as focused on the DL-based defect inspection methods proposed in 2013–2023 [16–22]. According to their summaries, the published methods can intuitively be divided into two stages with the gradual development of a CNN. The earlier stage is to combine CNNs and traditional ML pattern recognizers, such as [23], which uses the former to extract features and the latter to perform vision tasks.

Essentially, this mode still follows the solutions of feature engineering and thus fails to solve the problem of poor generalization. The latter stage is the end-to-end inspection methods, such as [24], which can perform inspection tasks on different kinds of surfaces, regardless of rigid or flexible ones. The methods proposed during this period are indeed effective, however, their successes are based on an implicit assumption: the training samples are enough. Unlike common objects, the collection of defect samples is very different in industry. If training data is limited, the CNNs may not converge and achieve excellent results. To alleviate the requirements for defect samples, the previous methods as in [25] reduced the size of a baseline CNN, which can also decrease the representation power of the model. The methods proposed in [26,27] adopted the pre-trained models for fine-tuning strong CNN models. Although this scheme can reduce the original sample requirement of the model, it cannot work well when there is a large gap between the target data distribution and source data distribution. Therefore, in the deep learning period, inspection methods pay more attention to solving the problem of lacking enough samples instead of how to design a model or an algorithm absorbedly, which is the fundamental difference from the traditional ML-based methods. Table 1 provides a concise summary of the reviewed research on surface defect detection based on deep learning.

Table 1. Summary of reviewed studies based on deep learning for surface defect inspection.

Reference	Topic	Publication Year
[16]	Surface defect detection using deep learning techniques and classify them into three categories: detection context, learning, and localization and classification.	2021
[17]	Discuss surface defect detection based on deep learning from three aspects: supervised method, unsupervised method, and weakly-supervised method.	2021
[22]	Steel surface defect recognition.	2022
[19]	A systematic review for vision-based defect recognition from a feature perspective.	2022
[18]	Deep learning defect detection methods for X-ray images.	2023
[21]	Deep CNN-based visual defect inspection.	2023
[20]	Summarize studies published between 2020 and 2023 in the field of deep learning-based surface defect detection applications in industrial products.	2024

For the data-limited defect inspection, the most intuitive solution is to generate new defect samples. Traditional data augmentation tricks, such as image rotations and flips, can simply expand the number of samples but cannot provide new knowledge, which is unable to assist a CNN in finishing feature learning under insufficient samples. In recent years, the sample generation paradigm has attracted much attention and provided a new solution for this problem. It can synthesize new samples by generative models, such as the Variational Auto-Encoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models (DMs). The literature [28–31] summarized the generative models and their applications in other fields. In 2022, He et al. [32] reviewed the defect detection methods based on GAN. Zhong et al. [33] investigated and analyzed the image generation of surface defects in 2023. The generative models are also in the domain of the DL. The above-mentioned literature [18–22] involved several defect inspection methods based on generative models, in which the papers were mainly published before 2022, but they did not focus on such methods. Therefore, this paper conducts a review of the generative models-based methods for the application of surface defect inspection that have been proposed in 2022–2024. In this paper, we categorized the popular generative models, i.e., the AE, GAN, and DM, and discussed their principles, properties, as well as advantages and disadvantages for industrial applications. Subsequently, we systematically summarized the related methods based on the three generative models. This paper also provides information about the common evaluation measures and public defect datasets. Finally, we discussed the key challenges and explored future work on the application of generative models in defect inspection.

The structure of this study is as follows: Section 2 outlines the popular generative models that can be used for inspection tasks. Section 3 presents a detailed overview of the generative model-based methods for surface defect inspection, where the papers are selected since 2022. Section 4 provides an introduction to the public datasets used in surface defect detection and the popular metrics used to evaluate surface defect detection models. The performance of the selected methods is also presented in this section. Section 5 concludes this study and offers a thorough discussion of the main challenges and future research directions for surface defect detection.

2. Overview of Generative Models

Conventional methods can generate defect images by using computer-aided design [34] or three-dimensional modeling [35]. However, this strategy has limitations: it requires additional ray-tracing, consumes extensive computational resources, and fails to generate defect objects with irregular shapes [32]. With the development of the architecture of deep neural networks, deep generative models are being widely used in defect inspection and gradually replacing the traditional methods. Generative models for surface defect inspection can be broken up into three kinds of network architectures: VAEs, GANs, and DMs, illustrated in Figure 2. Each architecture differs in the strategy that is used to learn a latent representation of the feature. Both VAEs and DMs learn to maximize the likelihood of the input data. In contrast, GANs utilize a minimax game between their two subnetworks, which can be implemented in any form of a differentiable system that maps input data from one space to another. In this section, we only briefly describe the working principle and recent application of these generative models. Readers can refer to the previous reviews and resources for more details on generative models, such as mathematical formulations.

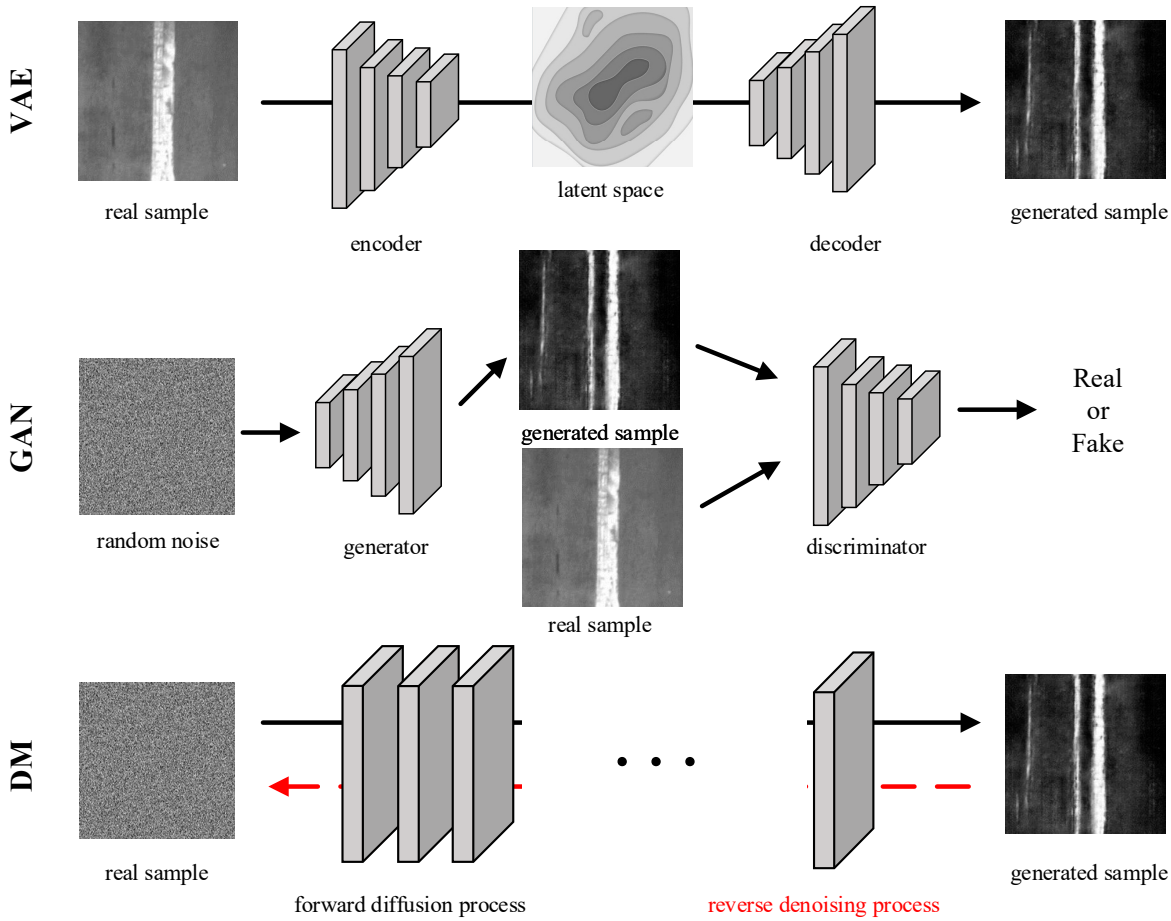

Figure 2. Comparison of major generative models used in defect inspection.

2.1. Variational Auto-Encoder

VAEs are generative models that intend to learn both an encoder and a decoder to map input data to values in a continuous latent space [28]. Consider defect generation: the encoder learns to map a defect sample into a continuous embedding that can be interpreted as a latent variable in a probabilistic model, while the decoder learns to reconstruct the defect sample from its learned embedding. The loss function of a VAE consists of two terms. One is the reconstruction loss term L_{rec} that forces the decoder to recover the original image x from its embedding $\hat{x}$. The other is the regularization term L_{reg} that aims to make the distribution of learned embedding $N(\mu, \sigma)$ conform to the standard normal distribution as much as possible, which measures the Kullback–Leibler (KL) divergence between these two distributions. The loss function is defined as in Equation (1).

$$L_{VAE} = L_{rec} + \lambda L_{reg} = ||x - \hat{x}||^2 + KL(N(\mu, \sigma), N(0, 1)) \tag{1}$$

where λ is the weight of the regularization term, which is used to balance the importance between the reconstruction loss and the regularization term. In the context of sample generation, VAEs can be used for signal generation [36,37], image generation [38,39], and text generation [40,41].

2.2. Generative Adversarial Network

GANs [29] are the deep generative models that consist of two neural networks: a generator G and a discriminator D. The G learns to create an image by sampling from a noise distribution p_z, i.e., transform the noise z into $G(z)$, with the aim to make the ideal p_z converge to the real data distribution p_{data}. Meanwhile, the D learns to distinguish whether the input sample x is real or fake. The optimization of a GAN can be viewed as a mini-max problem with value function V (G, D) that is written as in Equation (2).

$$\min_G \max_D V(G, D) = \mathbb{E}_{p_{data}(x)} \log D(x) + \mathbb{E}_{p_z(z)} \log[1 - D(G(z))] \tag{2}$$

where $\mathbb{E}$ is the empirical estimate of the expected value of the probability. In the optimization process, G and D compete with each other until reaching the Nash equilibrium [42]. At that point, G considers the generated samples to be realistic enough. GANs have been widely used in many fields, such as face recognition [43,44], video prediction [45,46], and natural language processing [47,48], etc. In the field of defect inspection, GANs are the most popular generative model to address the issues of data limitation, where more related details can be found in the next section.

2.3. Diffusion Model

DMs [30] are a family of probabilistic generative models that progressively destroy data by injecting noise and then learn to recover the data by reducing noise, in which the latter can be regarded as a sample generation process. Several methods make use of DMs for various tasks, such as super-resolution [49,50], image inpainting [51,52], and image translation [53,54], etc.

In computer vision tasks, the denoising diffusion probabilistic model (DDPM) is the commonly used diffusion model for image synthesis [55]. The DDPM can be viewed as a hierarchical Markov VAE with a fixed encoder. Specifically, the forward process of DDPM works as the encoder, but unlike VAE, this process is a linear Gaussian model. On the other hand, the reverse process of DDPM corresponds to the decoder shared between multiple decoding steps. Given that step t, $x_0 \sim q(x_0)$ represents the input data and its distribution, the forward process can be defined as the Markov chain $q(x_1, x_2, \ldots, x_T | x_0) = \prod_{t=1}^{T} q(x_t | x_{t-1})$ where x_t is the sample after injecting t-step noise to x_0. In contrast, for the prior distribution

p(x), the reverse process can be represented as $p_\theta(x_{t-1}|x_t) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$, where θ denotes model parameters. The loss function of DDPM is defined in Equation (3).

$$L_{DDPM}(x_0) = -\mathbb{E}_{q(x_0,x_1,\ldots,x_T)}[-\log p(x_T) - \sum_{t=1}^{T} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}] \quad (3)$$

3. Defect Inspection Based on Generative Models

In defect inspection, recent methods obtain generated samples through four types of generative models: VAEs, GANs, DMs, and multi-models, which combine with multiple models for sample generation. In addition, defect inspection tasks can be divided into three basic vision tasks: defect classification that identifies the defect class of a sample, defect detection that classifies and localizes each defect instance in a sample, and defect segmentation that divides a sample into regions with consistent characteristics pixel by pixel. In this section, we provide a technical review of the generative model-based defect inspection methods that are proposed between 2022 and 2024.

3.1. Defect Inspection Based on VAEs

VAEs have been widely used for image generation of surface defect inspection [32]. A simple way to use VAEs is for data augmentation, and then the generated samples are fed into a classifier or a detector. For instance, Xiang et al. [56] proposed the convolutional AE (CAE) for generating fabric defect samples and a defect detection algorithm based on Fourier convolution. Zhang et al. [57] designed the AE5-SSIM to reconstruct tinfoil images. This model is based on structural similarity reconstruction AE, whose encoder and decoder networks have five sub-encoder blocks and five sub-decoder blocks, respectively. Sae-ang et al. [58] introduced an auto-encoder to extract useful information from unlabeled data and then performed semantic anomaly detection on the samples. Si et al. [59] reconstructed images by a VAE model and analyzed the detection performance using various image gradient filters on the surface of wooden broads. Shi et al. [60] proposed a lightweight surface defect detection network based on the CAE to solve the problem of shortage of defective samples in the industry.

The data augmentation strategy is simple but limited application due to the generation and representation processes being separate. Two different learning stages may result in differences in feature learning and waste the opportunity to learn the correlation between them. Therefore, more researchers choose to establish defect inspection systems in end-to-end learning schemes. For VAE-based defect methods, Zhang et al. [61] proposed an attention-based vector quantization VAE (AVQ-VAE) for detecting defects in colored patterned fabrics, which adopts autoregressive modeling of discrete variables and attention mechanisms to enhance the detection performance. Zhang et al. [62] presented a multi-scale U-shaped denoising CAE (MSUDCAE) model for fabric defect detection, which was trained on defect-free samples and calculated residual maps between the original image and the corresponding reconstructed image to improve detection results. Tao et al. [63] presented an unsupervised detection method based on a dual-Siamese network. This framework tackled the problems that persist in the discriminative embedding-based methods and achieved state-of-the-art anomaly inspection performance on four public defect datasets. Min et al. [64] proposed a two-stage architecture including the railway cropping network and defects removal VAE (DR-VAE). The former extracted railroad surface and the latter carried out image reconstruction and defect detection on railway surfaces. Lu et al. [65] introduced a semantic label-enhanced (LE) VAE for generating defect examples and exploited semantic label information to improve classification performance. Fang et al. [66] proposed a sample-less anomaly detection strategy called FastRecon that works in low-data environments and is able to generalize across products at no additional cost. Li et al. [67] presented an introspective self-supervised reconstruction model (ISRM), which only used normal samples for training and integrated self-supervised learning into an introspective auto-

encoder to address the problems of sample imbalance and complex background interference. Yang et al. [68] established the Fs-CAE network by adding frequency shift operation based on the CAE network. The method only needs samples without defects for model training and does not require labels, thus reducing manual labeling time. Yang et al. [69] proposed an unsupervised defect detection method based on the Vision Transformer (ViT), which utilizes a transformer encoder to capture global contextual information, thereby enhancing detailed image understanding and detection performance.

Table 2 presents the key features of the selected papers that are based on VAEs for defect inspection. Due to the lack of defect images and manual annotations, these studies had to perform defect inspection in the modes of unsupervised learning or weakly-supervised learning. In 2022–2024, most methods focused on the challenging defect segmentation tasks. However, VAEs-based methods are only suitable for simple situations, for example, each image containing only one defect. VAEs generate samples by an encoding-decoding process, which assumes input samples follow a Gaussian distribution and maps them to the continuous data distribution [28]. Therefore, VAEs are good at reconstructing features with regular patterns but fail to learn complex and random features, resulting in the problem of generating blurry images [32]. Comparatively, GANs and DMs have more complex latent priors for data assumption, and thus have the ability to deal with a complex defect dataset [31].

Table 2. Summary of studies based on VAEs for surface defect inspection.

Method	Year	Brief Description	GM	Objective	Task	LM
[56]	2022	Generate image pairs with random masks for training.	FCAE	YDFID-1	CLS	USL
[57]	2022	Design the AE5-SSIM with five sub-encoders and five sub-decoders for tinfoil defect detection.	AE5-SSIM	tinfoil surface	CLS	USL
[58]	2022	Use pre-trained AE model for data augmentation and train Unet to predict the segmentation map.	VAE	MVTec	SEG	WSL
[59]	2023	Use a VAE for image reconstruction and image gradient filter for edge detection.	VAE	Wooden broad; MV-Tec	SEG	USL
[60]	2023	A lightweight reconstruction network based on CAE.	CAE	DAGM; MVTec; AITEX	SEG	USL
[61]	2022	Adopt autoregressive modelling of discrete variables to avoid the posterior collapse problem of traditional VAEs.	AVQ-VAE	YDFID-1; MVTec	SEG	USL
[62]	2022	A multi-scale U-shaped denoising CAE is modelled using defect-free samples.	MSUDCAE	MVTec; YDFID-1	SEG	USL
[63]	2022	Use a Siamese architecture to capture discriminative features of defect-free samples and obtain defect samples by a random generation module.	Dual-Siamese VAE	MVTec; KolektorSDD2; MT; RSDDs	CLS SEG	USL
[64]	2022	Combine supervised and self-supervised learning methods to complete the model training and detect defects on the rail surfaces automatically.	DR-VAE	Steel rail surface	SEG	USL
[65]	2023	Add semantic labeling information to improve interclass distance in potential space.	LEVAE	MVTec; Catenary	CLS	USL
[66]	2023	A few-shot anomaly detection strategy that works in a low-data regime.	FastRecon	MVTec	SEG	USL
[67]	2024	Integrate the concept of self-supervised learning into an introspective auto-encoder.	ISRM	RSDDs	CLS SEG	USL
[68]	2024	Add frequency shift operation on the basis of the CAE for surface defect detection.	Fs-CAE	air rudder surface	SEG	USL
[69]	2024	Employ ViT architecture that incorporates a transformer structure to understand the global context between image blocks.	ViT-based AE	MVTec	CLS SEG	USL

Notions: GM = generative model, CLS = classification, SEG = segmentation, LM = learning mode, USL = unsupervised learning, WSL = weakly-supervised learning.

3.2. Defect Inspection Based on GANs

GANs are the most popular generative model used in defect inspection. There are two main network architectures used for defect sample generation. The one is the classical GAN architecture that samples from the random noise distribution and generates samples unconditionally. Among the most representative models are standard GAN [42], DCGAN (deep convolutional GAN) [70], and WGAN (Wasserstein GAN) [71], etc. Existing inspection methods utilize these generative models for data augmentation or modify the output layer of the discriminator according to the type of inspection task. Table 3 summarizes the selected methods which are based on classical GAN architecture for defect inspection.

Table 3. Summary of studies based on typical GANs for surface defect inspection.

Method	Year	Brief Description	GM	Objective	Task	LM
[72]	2022	Utilize label assignment for generated samples and mix them into semi-supervised learning	Customized GAN	NEU-CLS-64; MT	CLS	WSL
[73]	2022	A triple attention semantic segmentation network for pixel-wise segmentation of surface defects	DA-GAN	DAGM; NEU-DET; MT	SEG	WSL
[74]	2022	Introduce latent variables in anomalous states to generate pseudo data	ALGAN	MVTec; MT	CLS	USL
[75]	2022	Use a typical GAN to generate leather samples	GAN	Leather	CLS	WSL
[76]	2022	Equip the GAN with fully convolutional networks and Markov discriminators	DefectGAN	Aluminum plate	CLS	WSL
[77]	2022	Classify defect based on WGAN and attention mechanism.	WGAN	X-SDD	CLS	WSL
[78]	2022	Construct GAN structure mapping the latent space to improve the quality of generated data	Latent Mapping GAN	Severstal	CLS	WSL
[79]	2023	Modify the DCGAN with ELU activation functions for sample generation	EDCGAN	NEU-CLS	CLS	WSL
[80]	2023	Make the generator cheat two discriminators simultaneously	GAN	NEU-CLS; aluminum profile	CLS	WSL
[81]	2023	Incorporate Vision Transformer into a GAN for image reconstruction	GAN with ViT	MVTec	CLS	WSL
[82]	2023	DCGAN and the encoder work together to produce a residual mapping with prominent defects.	DCGAN	Fabric	SEG	USL
[83]	2023	Incorporate both channel-wise and spatial-wise attention into GAN for sample generation.	ADRGAN	Hot-Rolled Steel Strip surface	CLS	WSL
[84]	2023	Simulate the background style of defect-free samples to obtain reasonable defects.	AGAN	KolektorSDD1; AITEX; RSDDs; MT	SEG	USL
[85]	2023	A contrastive learning framework for steel surface defect recognition with few labeled samples.	DGAN	NEU-CLS; NEU-CLS-64; X-SDD	CLS	WSL
[86]	2024	Employ two encoder–decoder models in GAN	DLS-GAN	Metal Iron Sheet	CLS DET	WSL
[87]	2024	The deployment time of the detection method is reduced by applying the Wasserstein distance and spectral normalized strategy	ACGAN	Conveyor belts.	CLS	WSL
[88]	2024	Use a pyramidal structure to learn the internal features of a single image.	PCSGAN	Railway surface.	DET	WSL
[89]	2024	Separate training on defects and backgrounds	defect-background separated GAN	Battery exterior; MVTec; KolektorSDD2	CLS SEG	WSL
[90]	2024	Use DCGAN to generate samples.	DCGAN	fabric stitching	CLS	USL

Notions: GM = generative model, CLS = classification, SEG = segmentation, LM = learning mode, USL = unsupervised learning, WSL = weakly-supervised learning.

For example, He et al. [72] designed a customized GAN to generate unlabeled samples and performed label assignment for unlabeled samples to enable them to participate in semi-supervised training. Liu et al. [73] introduced a triple-attention semantic segmentation network for surface defect detection, which utilized the GAN-based data augmentation (DA-GAN) method to exaggerate defect images and capture contextual information to improve segmentation results. Murase et al. [74] proposed the Anomalous Latent Variable Generating Adversarial Networks (ALGAN) that only used normal data for training and introduced latent variables in anomalous states to generate pseudo data. Gan et al. [75] used a GAN to generate pseudo-defect images and mixed them into real images in a pre-trained CNN for defect classification. For defect classification tasks, Li et al. [76] introduced the DefectGAN, which equipped fully convolutional networks and Markov discriminators to reduce the risk of pattern collapse and used a novel sampling strategy to balance memory consumption and training quality. Hao et al. [77] adopted the WGAN to generate new defect samples and integrated attention mechanisms into a CNN model to recognize defects on a hot-rolled strip steel surface. Song et al. [78] constructed a mapping network for the latent space, which made it possible to compose defect images of various sizes and optimized the proposed method in terms of convergence and computational cost. Jin et al. [79] replaced the activation functions DCGAN with ELUs (EDCGAN) for sample generation and adopted the MobileNet for steel surface defect classification. Liu et al. [80] modified the traditional GAN with two dual discriminators and introduced a regularization technique into GAN for generating high-quality and diverse surface defect images. Zhao et al. [81] presented a semi-supervised detection method of the metal product surface, which incorporated the Visual Transformer (ViT) into GAN and extracted multi-scale mask features for capturing rich semantic information. Wei et al. [82] proposed a multi-stage detection framework by transforming the defect detection task of fabrics into an unsupervised task based on DCGAN, which only required a few number of normal samples and achieved competitive results. Luo et al. [83] introduced the ADRGAN (attentive dual residual GAN) which uses the channel-wise and spatial-wise attention mechanisms to track and recover raindrop-like pseudo-defects. The method is able to locate and remove these pseudo-defects more accurately while retaining the real defect features, thus improving the accuracy and efficiency of defect detection. Ma et al. [84] constructed the adaptive GAN (AGAN) that generates defects similar in style to the background of the defect-free samples and designed the self-attentive encoder and spatial-aware decoder modules for obtaining the final prediction results. Xu et al. [85] proposed an unsupervised learning method for recognizing steel surface defects with fewer labeled samples, which generated defect images using a diverse generative adversarial network (DGAN) and learned the representation using flexible contrast to improve the semantic information of samples. Li et al. [86] proposed the Defect Location Sensitive GAN (DLS-GAN) for data augmentation, which used two encoder–decoders to help synthesize defects on the object and improve the synthesis quality. Wang et al. [87] modified the auxiliary classifier GAN (ACGAN) by applying the Wasserstein distance and spectral normalized strategy, which obtained high-quality generation and reduced the cost of detection. He et al. [88] proposed the perceptually constrained single image GAN (PCSGAN) to solve the problem posed by small numbers of defect images, which can train generators and discriminators at each scale. KU et al. [89] developed a defect-background separated GAN to enhance the accuracy of battery exterior defect inspection, which contributes to constructing a balanced dataset by increasing the overall quantity of the defect segmentation dataset, addressing the shortage of defect instances. Noor et al. [90] proposed a DCGAN-based method for the automatic generation of image synthesis of apparel stitching defects.

The other type of GAN architecture is based on image-to-image translation, which generates samples conditioned on input images, such as CycleGAN [91], StyleGAN [92], and SinGAN [93], etc. Using these generative models for constructing defect inspection methods can not only solve the issue of limited samples but also guide the learning mode of GANs due to the constraints on input samples. The selected methods that are

based on image-to-image translation GAN architecture are presented in Table 4. For instance, Niu et al. [94] introduced a weakly-supervised defect segmentation method that adopted the CycleGAN to generate defect-free templates through image-level annotation training and designed an attention module to enhance the ability to eliminate defective regions. Wen et al. [95] presented an unsupervised learning method suitable for defect detection with small samples, which expands the dataset by generating defective images and utilizes the attention mechanism to highlight key features in the images as a way to improve the accuracy of defect classification. Yi et al. [96] combined the SinGAN with the channel attention mechanism to generate high-quality and diverse defect images and improved classification accuracy by an image cutting and fusion method. Ren et al. [97] proposed an improved StyleGAN model that actively places different emphases on images of different resolutions to enhance the generation quality and employed different CNNs for classification. Hu et al. [98] proposed a structural defect detection algorithm based on the pix2pix network and migration learning using CGAN to reconstruct the defective region, calculate the difference between the reconstructed image and the defective image, and perform segmentation tasks on fabric surfaces. Wang et al. [99] introduced the foreground perception cycle-consistent adversarial network (FCGAN), which combines an attention network with a generator and a discriminator to recognize discriminative foregrounds and synthesizes coordinated and realistic pseudo-defect images. He et al. [100] designed a data augmentation module and a self-attention mechanism into the progressive GAN (Pro-GAN) to generate high-quality and high-diversity surface defect images. Xu et al. [101] combined dual U-Net networks with the pix2pix network to generate defect localization maps, and then employed the YOLOv3 for fabric defect detection. Ran et al. [102] proposed a new GAN for image-to-image translation to achieve high-quality synthesis from masked images to defect images of steel strips and validated the effectiveness of the synthesized data for data augmentation in YOLOv8. He et al. [103] proposed the production-and-elimination GAN (PE-GAN) for high-quality defect sample generation, which guides the learning direction of generative models by simulating the formation of defects on the surface of steel plates.

Table 4. Summary of studies based on image-to-image translation GANs for surface defect inspection.

Method	Year	Brief Description	GM	Objective	Task	LM
[94]	2022	Use CycleGAN to generate dynamic templates with image-level annotations	CycleGAN	DAGM	SEG	WSL
[95]	2022	Combine the CycleGAN with attention mechanism for synthesizing defect samples to enlarge the samples volume	CycleGAN	DAGM; KolektorSDD1	CLS	USL
[96]	2023	Integrate channel attention mechanism into SinGAN	SinGAN	NEU-DET	CLS	WSL
[97]	2023	Introduce style-related features to enhance the control on the generated images	StyleGAN	Ceramic surface	CLS	WSL
[98]	2023	Integrate self-attention mechanism into the pix2pix model for reconstructing defect regions with less information	pix2pix	Fabric	SEG	USL
[99]	2023	Incorporate attention network to guide the training process	FCGAN.	KolektorSDD1; DAGM	CLS	WSL
[100]	2023	Integrate data augmentation and self-attention mechanism modules into Pro-GAN	Pro-GAN	NEU-DET	CLS DET	WSL
[101]	2024	Use dual U-net networks to enhance defect-texture fusion	Pix2Pix	Fabric	DET	WSL
[102]	2024	Extract defect edges of original image as well as the background texture information, and use them as network input through the spatially-adaptive normalization module	pix2pix	Severstal; MT	DET SEG	WSL
[103]	2024	Design a production-and-elimination process to improve generation quality	PE-GAN	NEU-CLS-64	CLS	WSL

Notions: GM = generative model, CLS = classification, SEG = segmentation, LM = learning mode, USL = unsupervised learning, WSL = weakly-supervised learning.

3.3. Defect Inspection Based on DMs

Although it has been proven that the generation quality of DMs outperforms that of GANs and VAEs, the manufacturing field seems to pay little attention to DM-based defect inspection and sample generation. Tang et al. [104] presented an unsupervised learning defect detection method on fabric surfaces, which improves the quality of image generation by optimizing the time step and utilizes the simplex noise DDPM (SN-DDPM) to reconstruct defect images, thus realizing accurate segmentation of defects. Huang et al. [105] proposed a hybrid DL framework based on DDPM for data enhancement and a deep residual neural network (DenseNet) for defect classification on composite panel surfaces. Tao et al. [106] generated diverse defective images by training a DDPM model with limited samples for generalized surface defect inspection. In this method, an erasing–inpainting mechanism is proposed to promote the recombination of real features and learned features, thus fully using the limited defect samples. Liang et al. [107] introduced a stabilized diffusion enhancement method based on the improved YOLOv5 model for defect detection in metal stamping parts. This method introduces a lightweight bottleneck transformer structure and a multi-head self-attention mechanism to increase detection speed and accuracy.

Table 5 summarizes the DM-based studies conducted for surface defect inspection. Although DMs have strong performance and can generate realistic images, the network structure of DMs is too heavy and the sampling time is too long, which requires a significant amount of computation resources and training time. There is room to reduce training difficulty and model complexity for DMs, resulting in applicability for industrial applications in the future.

Table 5. Summary of studies based on DMs for surface defect inspection.

Method	Year	Brief Description	GM	Objective	Task	LM
[104]	2023	Employ the defect-free samples to train model with time-step adaptive modules	SN-DDPM	YDFID-1	SEG	USL
[105]	2023	Use DDPM for data enhancement and DenseNet for defect classification	DDPM	Composite panel surface	CLS	WSL
[106]	2024	Train a DDPM model using limited defective images and use an erasing–inpainting operation to improve generation quality	DDPM	NEU-CLS	CLS SEG	WSL
[107]	2024	Use stable diffusion improvement to augment the dataset	DDPM	Metal Stamping part surface	DET	WSL

Notions: GM = generative model, CLS = classification, SEG = segmentation, LM = learning mode, USL = unsupervised learning, WSL = weakly-supervised learning.

3.4. Defect Inspection Based on Multi-Models

A new trend builds generative models based on multiple models, e.g., the combination of a GAN and a VAE. Zhao et al. [108] introduced the channel attention mechanism into AE generator network and Combine with the Wasserstein GAN (WGAN) for image restoration, which can achieve good results in removing the oil droplets from the image and restoring the texture information of the oil droplet region. Yang et al. [109] introduced the Mask2Defect that utilizes priori knowledge to data augmentation and has the ability to use generated samples to fine-tune the inspection model. The Mask2Defect consists of three generative models: the IPK-Encoder (industrial prior knowledge encoder) that obtains teacher masks for semi-supervised training, the M2DCNet (mask-to-defect construction network) that renders the defect details based on teacher masks, and the F2RDT-GAN (fake-to-real domain transformation GAN) that transforms the rendered samples to real defect domain for guaranteeing realistic background textures. Yu et al. [110] proposed a weakly-supervised learning method for fabric defect detection, which replaces the generator of GAN with an AE architecture to generate samples. Then, the single-shot multi-box detector network is adopted for multi-scale defect detection. Ferrari et al. [111] introduced an

anomaly detection architecture that combines a GAN and a residual AE, which improves the accuracy of detecting and localizing surface anomalies by focusing on specific regions of the image. Dai et al. [112] combined an attention feature fusion module and a residual banalization module through an encoder–decoder–encoder network structure to perform defect detection and segmentation based on defect-free samples. Bu et al. [113] combined the deep conditional VAE (DCCVAE) and auxiliary conditional Wasserstein GAN with gradient penalty (ACWGAN-GP) to gradually expand and generate various coating defect images for solving the overfitting problem due to unbalanced data and finally adopted a CNN for defect classification based on generated samples.

Table 6 presents a summary of selected papers that combine multiple generative models for surface defect inspection. The commonly used combination is to integrate a VAE and a GAN, which can combine their advantages and achieve outperformance.

Table 6. Summary of studies based on multi-models for surface defect inspection.

Method	Year	Brief Description	GM	Objective	Task	LM
[108]	2022	Introduce channel attention mechanism to improve detail recovery ability in the raindrop removal area	AE-WGAN	Bearing	CLS	WSL
[109]	2022	Fuse prior knowledge data to generate defects with varied features and regard the generated defects as teacher samples to fine-tune the inspection model	Mask2Defect	Automobile parts; MT	CLS SEG	WSL
[110]	2023	Design an improved GAN with an AE architecture to address the paucity of requisite defective samples	GAN	Fabric	CLS	WSL
[111]	2023	Combine a GAN and a residual AE for defect segmentation in semi-supervised learning manner	GRD-Net	MVTec	CLS SEG	WSL
[112]	2023	Build a network model based on encoder–decoder–encoder and incorporate an attention feature fusion module in the decoder	F-GAN	MVTec	CLS	USL
[113]	2024	Combine DCCVAE and ACWGAN-GP for image generation and recognition of ship coating defects	DCCVAE-ACWGAN-GP	Ship coating	CLS	USL

Notions: GM = generative model, CLS = classification, SEG = segmentation, LM = learning mode, USL = unsupervised learning, WSL = weakly-supervised learning.

4. Datasets and Metrics

In this section, we provide a summary of public industrial defect datasets that are widely used for surface defect inspection. Then, we introduce the standard metrics for the industrial inspection methods that are based on generative models. Finally, by these metrics, the comparison results obtained from the selected papers are presented for the public defect datasets.

4.1. Surface Defect Datasets

In the field of industry, a dataset plays a crucial role in the research of generative model-based defect inspection methods. Unlike common detection datasets, defect datasets have smaller sample sizes with fewer categories and prefer to use grayscale images. According to the summary of the selected methods in the last section, the extensively used defect datasets are summarized in Table 7, specifically as follows: AITEX [114], DAGM [115], KolektorSDD1 [116], KolektorSDD2 [117], MT (Magnetic Tile) [118], MVTec [119], NEU series (NEU-CLS [120], NEU-DET [24], and NEU-CLS-64 [72]), RSDDs [121], X-SDD [122], and YDFID-1 [123]. These datasets cover a wide range of defect inspection scenarios of manufacturing product surfaces.

Table 7. Summary of public surface defect datasets.

Dataset	Objective	Description	
AITEX [114]	Fabric	Number of defect-free images: 140 Number of defect images: 105	Number of defect types: 12 Size of image: 4096 × 256
DAGM [115]	Artificial defect	Number of defect-free images: 6000 Number of defect images: 1500 Number of defect types: 10	Size of image: 512 × 512 With oval annotations
KolektorSDD1 [116]	Metal Surface	Number of defect-free images: 347 Number of defect images: 52	Number of defect types: 1 Size of image: 1240–1270 × 500
KolektorSDD2 [117]	Metal Surface	Number of defect-free images: 2979 Number of defect images: 356 Number of defect types: 3	Size of image: 230 × 630 With segmentation annotations
MT [118]	Magnetic tile	Number of defect images: 1344 Number of defect types: 6	Size of image: 128 × 128 With saliency annotations
MVTec [119]	Multi-type	Number of defect images: 5354 Number of defect types: 15	Size of image: 700 × 700–1024 × 1024
NEU-CLS [120]	Steel plate	Number of defect images: 1800 Number of defect types: 6	Size of image: 200 × 200
NEU-DET [24]	Steel plate	Number of defect images: 1800 Number of defect types: 6	Size of image: 200 × 200 With bounding-box annotations
NEU-CLS-64 [72]	Steel plate	Number of defect images: 7000 Number of defect types: 9	Size of image: 64 × 64
RSDDs [121]	Rail surface	Number of defect images: 195 Number of defect types: 2	Size of image: 1500 × 2180
X-SDD [122]	Hot-rolled strip	Number of defect images: 1360 Number of defect types: 7	Size of image: 128 × 128
YDFID-1 [123]	Fabric	Number of defect-free images: 3189 Number of defect images: 312	Number of defect types: 3 Size of image: 512 × 512

4.2. Metrics

Metrics are very important and necessary which can guide model design and demonstrate the effectiveness and superiority of the proposed methods. For surface inspection, it is desirable first to evaluate the accuracy of the inspection methods. The most basic indicators are accuracy (ACC), precision, recall, and F1-score, where the definitions of these indicators are defined in Equations (4)–(7).

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

$$F1 - score = \frac{2 \times Recall \times Presion}{Recall + Presion} \tag{7}$$

where TP, TN, FP, and FN indicate a situation in which the model correctly predicts a positive sample as a positive case, the model correctly predicts a negative sample as a negative case, the model incorrectly predicts a negative sample as a positive case, and the model incorrectly predicts a positive sample as a negative case, respectively.

Additionally, in order to reasonably evaluate the prediction effect of methods, the indicators mAP (mean Average Precision) and mIoU (mean Intersection over Union) are always

used for surface defect detection and segmentation tasks. For k-class defect inspection, they are calculated as in Equations (8) and (9).

$$mAP = \frac{1}{k}\sum_{i=1}^{k}\frac{Recall + Presion}{2} \tag{8}$$

$$mIoU = \frac{1}{k}\sum_{i=1}^{k}\frac{|X \cap Y|}{|X \cup Y|} \tag{9}$$

where X denotes the predictions and Y denotes the ground-truths. Similar to mAP, AUROC is also a common indicator for surface defect inspection. The AUROC means the area under the receiver operating characteristic (ROC) curve, where the calculation formula is as Equation (10).

$$AUROC = \frac{\sum_{S_i \in positiveclass} rank_{s_i} - \frac{M \times (M+1)}{2}}{M \times N} \tag{10}$$

where $\sum_{S_i \in positiveclass}$ represents the summation of the order number of positive samples. $rank_{s_i}$ refers to rank the positive samples according to the probability scores. M and N represent the order number of positive and negative samples, respectively. There are two kinds of AUROC indicators: the image-level AUROC for evaluating classification performance and the pixel-level AUROC for evaluating segmentation performance.

The generated samples are fake images synthesized by generative models, whose quality can greatly affect the performance of inspection models. The Fréchet Inception Distance (FID) is the popular metric for evaluating the generation quality by comparing the feature distribution difference between generated samples and real samples. The definition of FID is shown in Equation (11).

$$FID(x, x\prime) = \left|\left|\mu_x - \mu_{x\prime}\right|\right| + Tr(\Sigma_x + \Sigma_{x\prime} - 2\sqrt{\Sigma_x \Sigma_{x\prime}}) \tag{11}$$

where x and $x\prime$ are the real and the generated samples, respectively. The notions μ, Tr, and Σ represent the mean, the covariance matrix, and the trace of a matrix.

4.3. Comparisons on Public Defect Datasets

Table 8 summarizes the comparison results of selected methods on the public defect datasets which have been mentioned in Section 4.1. Specifically, the comparison performance of the AITEX, DAGM, KolektorSDD1, KolektorSDD2, MT, MVTec, NEU-CLS, NEU-DET, NEU-CLS-64, RSDDs, X-SDD, and YDFID-1 datasets are shown in Table 8 (a)–(k).

Overall, the surface inspection methods based on generative models have made significant achievements since 2022, especially the defect classification tasks. However, the defect detection and segmentation performance still face challenges due to instable generation quality and having to train models in unsupervised learning or weakly-supervised learning manners. It is worth noting that most inspection methods do not open source code for its potential for industrial applications in the future, and thus it is impossible to reproduce them.

Table 8. Comparison results of selected methods on the public defect datasets.

(a) AITEX

Method	Recall (%)	Precision (%)	F1-Score (%)	mIoU (%)	mAP (%)
[60]	88.4	77.2	82.4	-	-
[84]	-	-	-	89.61	91.46

(b) DAGM

Method	Recall (%)	Precision (%)	F1-Score (%)	mIoU (%)	FID
[60]	94.8	93.5	94.1	-	-
[73]	96.4	98.1	97.2	86.9	-
[94]	-	-	-	78.28	-
[95]	-	-	-	-	168
[99]	-	-	-	-	93.83

(c) KolektorSDD1

Method	ACC (%)	mIoU (%)	mAP (%)	FID
[84]	-	88.91	89.55	-
[95]	98.53	-	-	122
[99]	99.8	-	-	115

(d) KolektorSDD2

Method	ACC (%)	mIoU(%)	AUROC (%) (image-level)	AUROC (%) (pixel-level)
[84]	-	88.91	89.55	-
[95]	98.53	-	-	122
[99]	99.8	-	-	115

(e) MT

Method	ACC (%)	Recall (%)	Precision (%)	F1-score (%)	mIoU (%)	mAP (%)	AUROC (%) (image-level)
[63]	-	-	-	-	-	-	89.1
[72]	99.22	-	-	-	-	-	-
[73]	-	94.9	96.3	95.6	85.1	-	-
[74]	-	-	-	-		-	95.6
[84]	-	-	-	-	88.36	90.32	-
[102]	-	81.2	98	-	-	82.5	-
[109]	89.32	-	-	-	83.7	-	-

(f) MVTec

Method	ACC (%)	F1-score (%)	mIoU (%)	AUROC (%) (image-level)	AUROC (%) (pixel-level)
[58]	-	-	66.6	-	-
[59]	-	93.1	-	-	-
[60]	-	88.1	-	-	-
[61]	-	-	69	84	-
[62]	-	-	-	71.7	-
[63]	-	-	-	96.1	96.7

(g) NEU-CLS/NEU-DET

Method	ACC (%)	Recall (%)	Precision (%)	F1-score (%)	mIoU (%)	mAP (%)	FID
[73]	-	96.1	97.2	96.6	86.3	-	-
[79]	95.74	-	-	-	-	-	-
[80]	-	-	-	-	-	72.26	21.74
[85]	96.47	-	-	-	-	-	152.36
[96]	94.67	94.24	94.36	94.3	-	-	-
[100]	-	-	-	-	-	72.1	33.4
[106]	99.67	96.8	87.4	99.7	-	92.3	104.14

Table 8. *Cont.*

(h) NEU-CLS-64					
Method	**ACC (%)**		**FID**		
[72]	99.38		-		
[85]	88.28		-		
[103]	99.79		153.15		
(i) RSDDs					
Method	**mIoU (%)**	**mAP (%)**	**AUROC (%) (image-level)**	**AUROC (%) (pixel-level)**	
[63]	-	-	100	98.8	
[67]	-	-	98.5	97.2	
[84]	86.89	87.88	-	-	
(j) X-SDD					
Method	**ACC (%)**	**Recall (%)**	**Precision (%)**	**F1-score (%)**	
[77]	99.2	99.21	99.29	99.24	
[85]	86.4	-	-	-	
(k) YDFID-1					
Method	**ACC (%)**	**Recall (%)**	**Precision (%)**	**F1-score (%)**	**mIoU (%)**
[56]	92.4	80.5	74.2	77.2	-
[61]	99.28	58.71	64.41	53.87	37.44
[62]	98.42	-	-	-	50.46
[104]	98.95	-	-	58.52	46.63

5. Discussion and Conclusions

Generative models maintain rapid development as a powerful tool for defect inspection on manufacturing product surfaces. In this paper, we have conducted a comprehensive review of the recent defect inspection methods that were proposed between 2022 and 2024. Specifically, we focus on the generative models-based methods because there is generally a shortage of defect samples in the industrial field. Furthermore, we introduce the public defect datasets and the comparison results of the selected methods. Finally, we discuss the challenges that still need to be addressed in this area. The goal of this review is to offer readers a greater understanding of the present research status on surface defect inspection for industrial products, to serve as a reference for future research in this area.

5.1. Challenges and Opportunities

Although generative models have been widely used in the field of defect inspection, there are some challenges with both the model itself and its application to the task of sample generation of surface defects.

- Small-scale defect datasets. Due to the difficulty in collecting defect samples, existing industrial surface defect datasets are relatively small scale. For deep neural networks, the larger model size brings stronger representability that can achieve better inspection results. However, the model size is positively correlated with sample requirements. A small training set limits the application of large models in industrial inspection tasks. Therefore, it is very cheap to obtain samples through generative models, which have been the research hotspot in the field of industrial inspection.Not only in terms of the number of defect samples, manual annotations are also insufficient due to the lack of simple and reliable labeling methods [72]. Undoubtedly, inspection methods can achieve excellent accuracy under the supervision of manual labels. However, industrial labeling is highly professional and heavy work, resulting in most existing defect datasets only having image-level labels. Meanwhile, the generated samples are unlabeled, and inspection models can hardly be constructed in a supervised learning manner with these samples. Therefore, current methods can only utilize generated

samples for defect inspection tasks, especially for defect detection and segmentation tasks, in weakly-supervised learning and unsupervised learning manners. The accuracy obtained in these ways cannot meet the actual levels of industrial inspection.Furthermore, the small scale of a dataset can lead to an imbalance in representation learning during the process of sample generation [124]. For large-scale datasets, it is not necessary to require equal sample sizes between classes, which means class imbalance can slightly affect the training of deep models. However, defect datasets are all small-scale, in which the number of defect-free samples is usually bigger than that of defect samples. If training on such a class-imbalanced dataset, the model tends to learn features in features of categories with larger sample sizes. Although it is easy to recover the balance of a dataset by adding generated samples, these samples are essentially fake images, which may not necessarily have a positive impact on generation learning.

- Difficulties in training of generative models. Advanced generative models have their own difficulties in training. The goal of training a GAN is to reach Nash equilibrium, but it is difficult to achieve because the adversarial nature of a GAN cannot guarantee convergence, and thus training usually occurs in oscillations with gradually increasing amplitudes [125]. Moreover, GANs may suffer from mode collapse, which causes the generator to only learn simple features and fail to learn complex features, and thus GANs only produce duplicate or similar images [29]. For the DMs, the sampling time as well as the training time are too long, and their model structures are too complex and heavy. Moreover, generative models are also deep neural networks, which require a certain number of samples for model training. The aim of generative models is to generate images as realistically as possible. Therefore, generative models need to train on real samples, which in some cases cannot be achieved when the number of initial defect samples is extremely small. A method that allows training models on a small number of samples urgently needs to be developed.
- How to improve generation quality. The performance of inspection models is highly dependent on the quality of generation samples. The quality not only refers to the clarity of the generated image but also to the diversity of generated defect features. It is necessary to design a learning mechanism to guide generative models to focus on defect areas and ignore background areas. Moreover, many product surfaces can produce defects of small size. For example, a defect target only occupies 32×32 pixels area in a 512×512 pixels defect image. Generative models are essentially deep neural networks in which each convolutional layer is equivalent to a filter. It is very challenging for generative models to learn and generate such a small target that only has a few details. Feature fusion techniques can bring more details to feature representation, which boosts the recognition of small defect targets [24]. However, this method is not always effective. A better method for discovering target details needs to be studied for the further improvement of the accuracy of detecting small defect targets. The background texture can also interfere with the generation quality. Many semi-finished products without surface treatment, e.g., hot-rolled steel plates, also need surface defect inspection to ensure production quality. Sometimes background textures may confuse a detector's cognition and be identified as defects, while subtle defects may be identified as background textures. In addition, many industrial production environments are extremely harsh, resulting in oil contamination or dust appearing on the surface of the product. These objects are generally not included in defect datasets and can therefore be misclassified as existing defect classes in the dataset, causing generative models to perform incorrect feature learning.
- Real-time inspection. The primary objective of industrial inspection is undoubtedly the pursuit of high precision. The design of defect inspection methods should also satisfy the requirements of industrial real-time performance. However, the reality is current methods, especially based on deep learning, are involved in the trade-off of inspection accuracy and speed. Although computer hardware has developed rapidly

and model lightweight technology has made some progress, accuracy and real-time performance still cannot be balanced for industrial inspection.

5.2. Future Research Directions

- A generative model with high generation quality. Generative models are not intended as a means of data augmentation. If the generated samples are not clear enough, it will have a negative impact on feature learning. For defect inspection methods based on generative models, a training set always contains a much larger number of generated samples than real ones. Only increasing the number of samples cannot improve the final detection accuracy. DM is currently the most advanced and powerful generative model, which holds great potential in defect detection. However, this model still needs improvements due to difficulties in training, slow sampling speed, and other reasons. An important research direction will be to solve the problems of DMs and design inspection methods based on these models.
- Knowledge transferring. The distribution differences exist between the generated sample and the real sample. Due to the difficulty in obtaining new samples, the differences can seriously affect the quality of generation and inspection results. A promising research direction is to mine common knowledge and suppress special knowledge across different defect datasets. For example, both steel and aluminum plates have scratch defects that have similar features. The scratch features are common knowledge, and correspondingly different surface textures are their special knowledge. If aluminum plate datasets have a few samples, common knowledge from steel plate datasets can be transferred to boost sample generation. The most commonly used knowledge-transferring method is the cross-domain adaptation [126]. The effectiveness of mining knowledge from fake data is disappointing, which is a research problem that needs to be investigated.
- Accurate defect detection or segmentation in the weakly-supervised learning mode. Due to generated samples being unlabeled, most defect inspection methods are in the weakly-supervised learning mode. Although effective at defect classification, the detection and segmentation accuracy is far inferior to supervised learning methods. On one hand, it is because of the limitations of weakly-supervised learning itself; on the other hand, it is due to the lack of reliable label assignment methods, which are worth studying in future work.
- Lightweight model design. There are successful lightweight design cases in convolutional neural networks, e.g., the MobileNet [127]. However, these schemes are unable to integrate into generative models. To obtain high generation quality, powerful generative models require a complex learning process, such as the adversarial learning of GANs and the diffusion-and-denoising process of DMs, which depends on a heavy network architecture. Finally, the generation process is always time consuming. To meet real-time inspecting, generative models require lightweight design, which can also reduce the sample demand.

Author Contributions: Conceptualization, Y.H. and X.W.; methodology, Y.H. and J.X.; software, Y.H. and S.L.; validation, S.L. and X.W.; investigation, S.L.; resources, Y.H.; data curation X.W.; writing—original draft preparation, Y.H.; writing—review and editing, X.W. and J.X.; visualization, Y.H.; project administration, Y.H. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the National Natural Science Foundation of China (62306186), the Technology Plan Joint Foundation of Liaoning Province (2023-BSBA-238), and the Technology Plan Joint Foundation of Liaoning Province (2023-MSLH-246).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflicts of interest.

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